EE3355: SIGNALS AND SYSTEMS

LABORATORY 01

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Table 0-1: Summative Laboratory Form

Semester	03
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Module Name	Signals and Systems
Lab Number	01
Lab Name	Continuous-time Signal Analysis
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1 Observations

1.1 Part 01

1.1.1 Output of the sinusoidal waveform

Table 1-1: Properties of sinusoidal waveform in time domain

Property	Value
Amplitude	1.42 mV
Frequency	50.27 Hz
Period	19.89 Ms

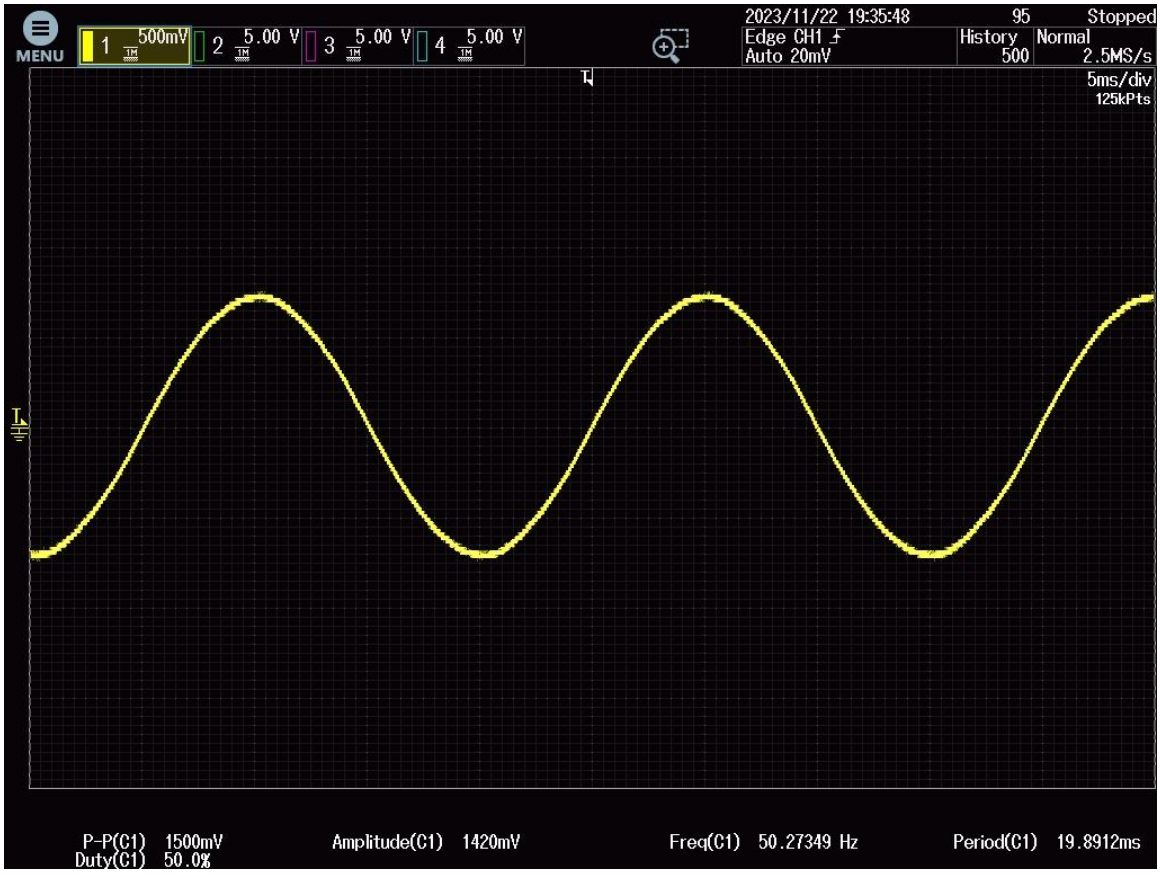


Figure 1-1: Sine waveform in time domain

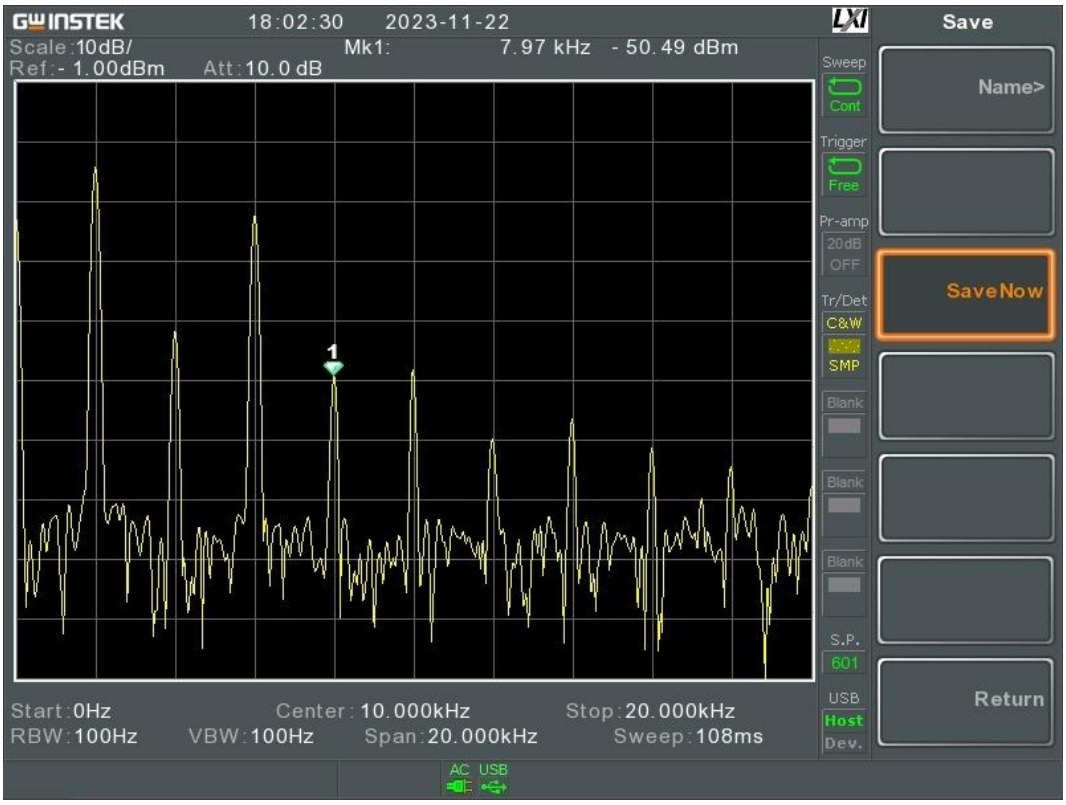


Figure 1-2: Sine wave in the frequency domain observed by spectrum analyzer

1.1.3 Observations of the square waveform



Figure 1-3: Square wave in Time domain with 10% duty cycle

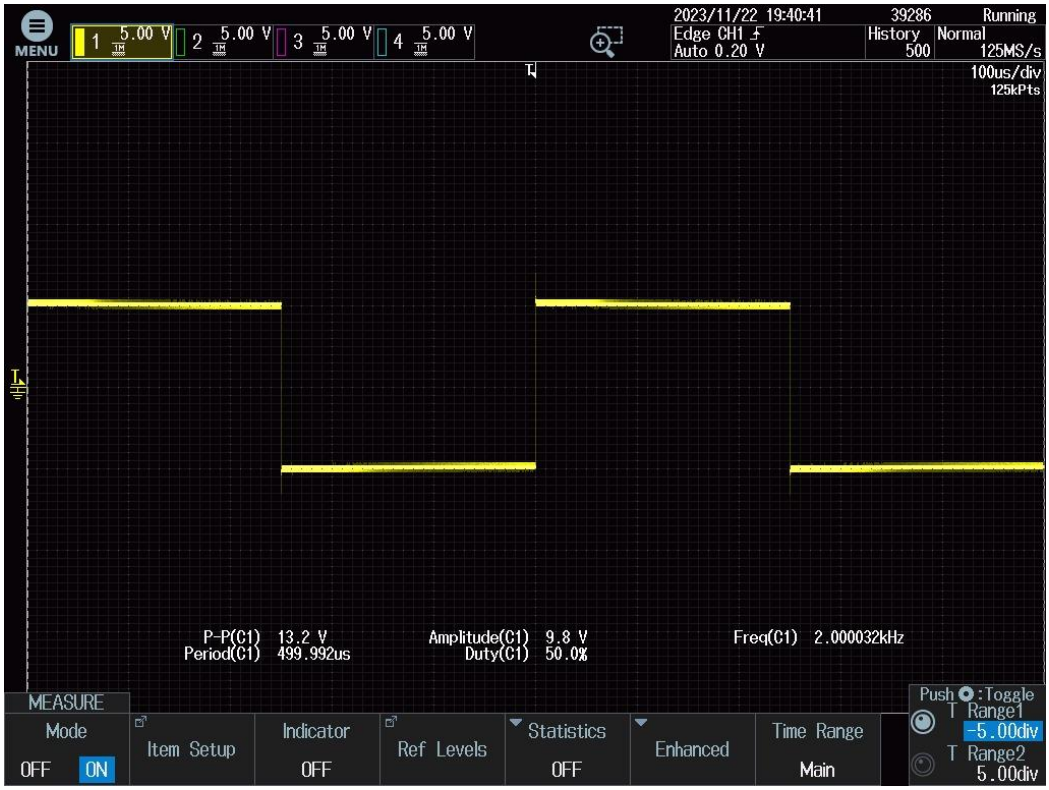


Figure 1-4: Square waveform in time domain with 50% duty cycle



Figure 1-5: Sine waveform in time domain with 90% duty cycle

1.2 Part 2

1.2.1 Lissajous Pattern for $f_y = 2f_x$

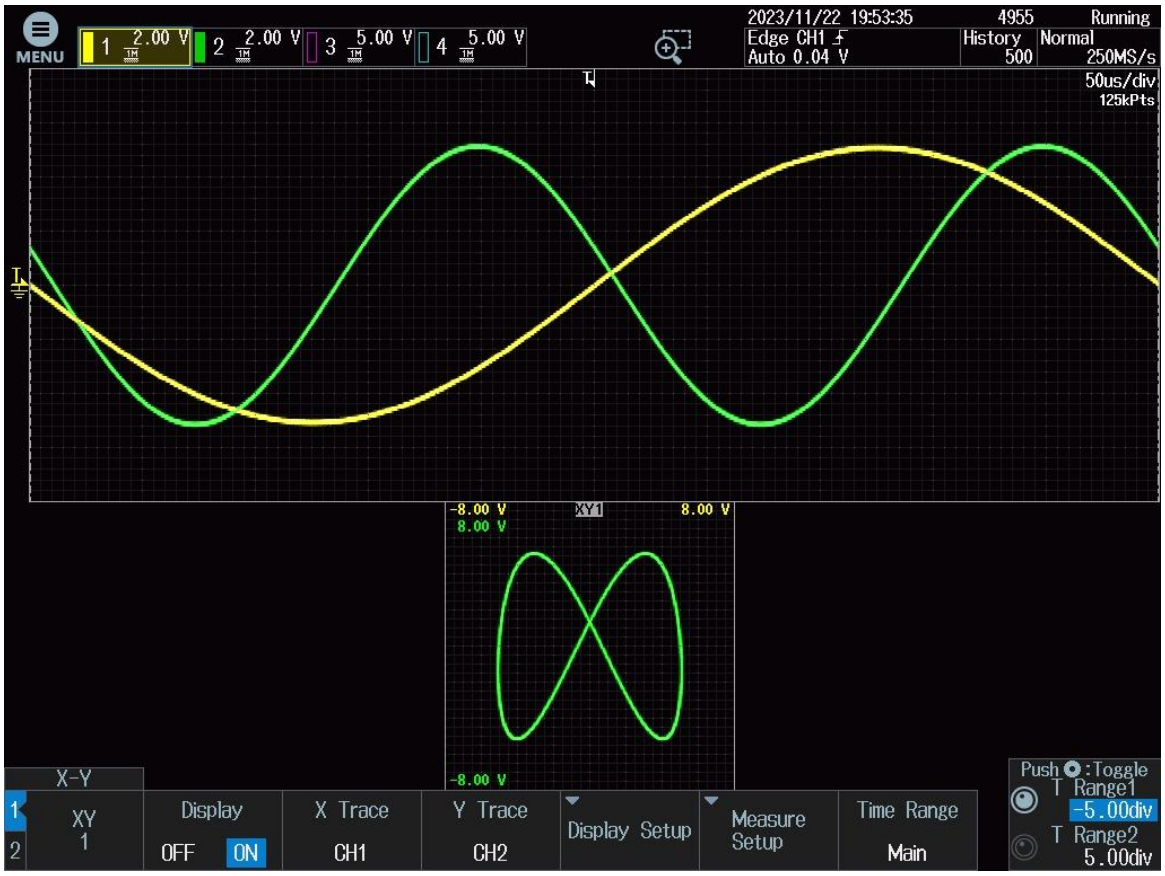


Figure 2.1: Lissajous pattern for two waveforms with amplitude ration 1:2 90 degree phase shift

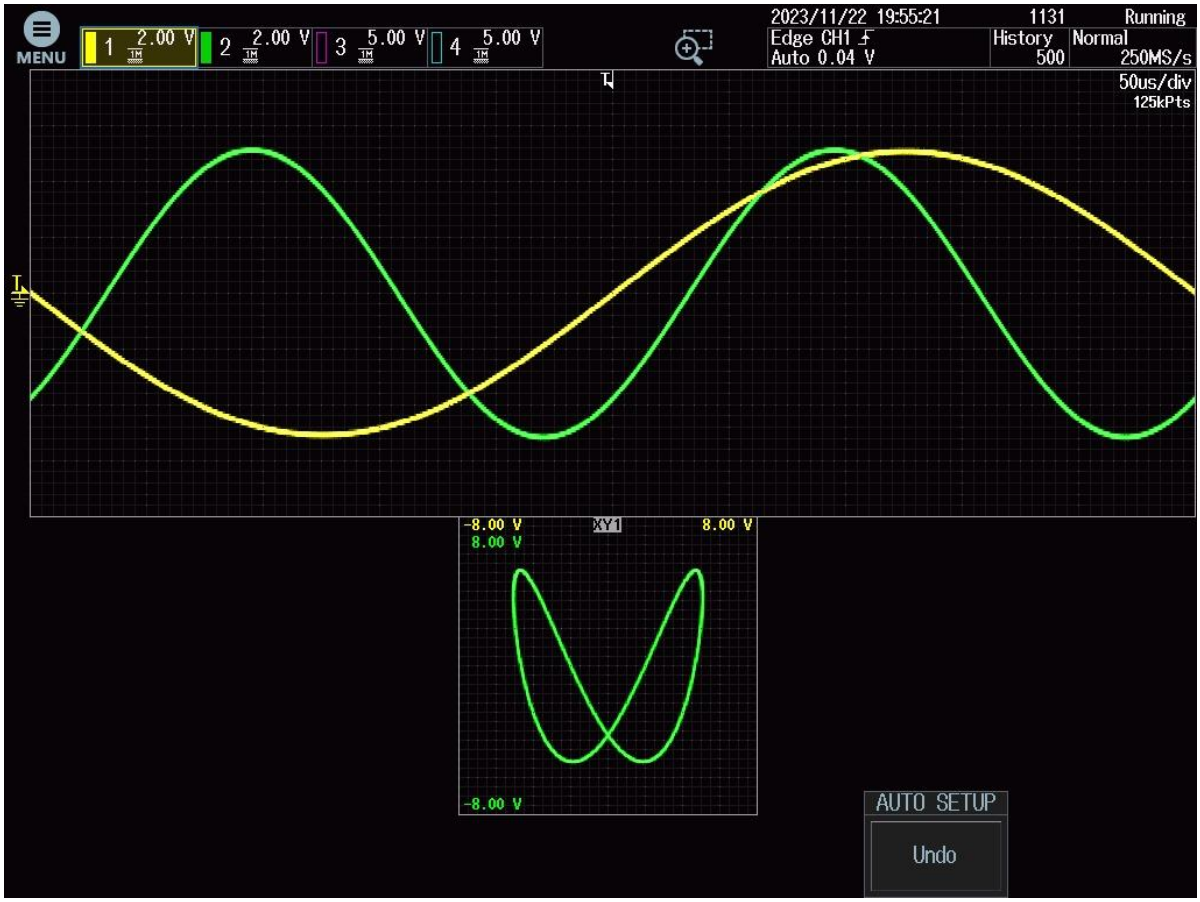


Figure 2.2: Lissajous pattern for two waveforms with amplitude ratio 1:2 with 45-degree phase shift

2 Calculations

The frequency ratio of a Lissajous pattern is found by drawing vertical and horizontal lines through the pattern's center. The frequency ratio is determined by dividing the number of horizontal axis crossings by the number of vertical axis crossings. That is,

$$Frequency\ ration = \frac{number\ of\ horizonatl\ crossing}{number\ of\ vertical\ crossing}$$

The Lissajous pattern in Figure 2.1 is generated by two sinusoidal waveforms, one with twice the voltage.

Considering the Lissajous pattern for Figure 2.1,

$$\begin{aligned} Frequency\ ration &= \frac{f_y}{f_x} \\ &= \frac{4}{2} \\ &= 2 \end{aligned}$$

3 Discussion

1. Phase angle and magnitude relationships of two sinusoids of the Lissajous pattern

The Lissajous pattern is a geometric shape generated by two sinusoids intersecting. The shape of the Lissajous pattern is determined by the phase angle and magnitude relationships of the two sinusoids. They are created by applying two alternating current waveforms to an oscilloscope's horizontal and vertical amplifiers. If one of the waves' information is known, these patterns can be used to identify the information of the second waveform.

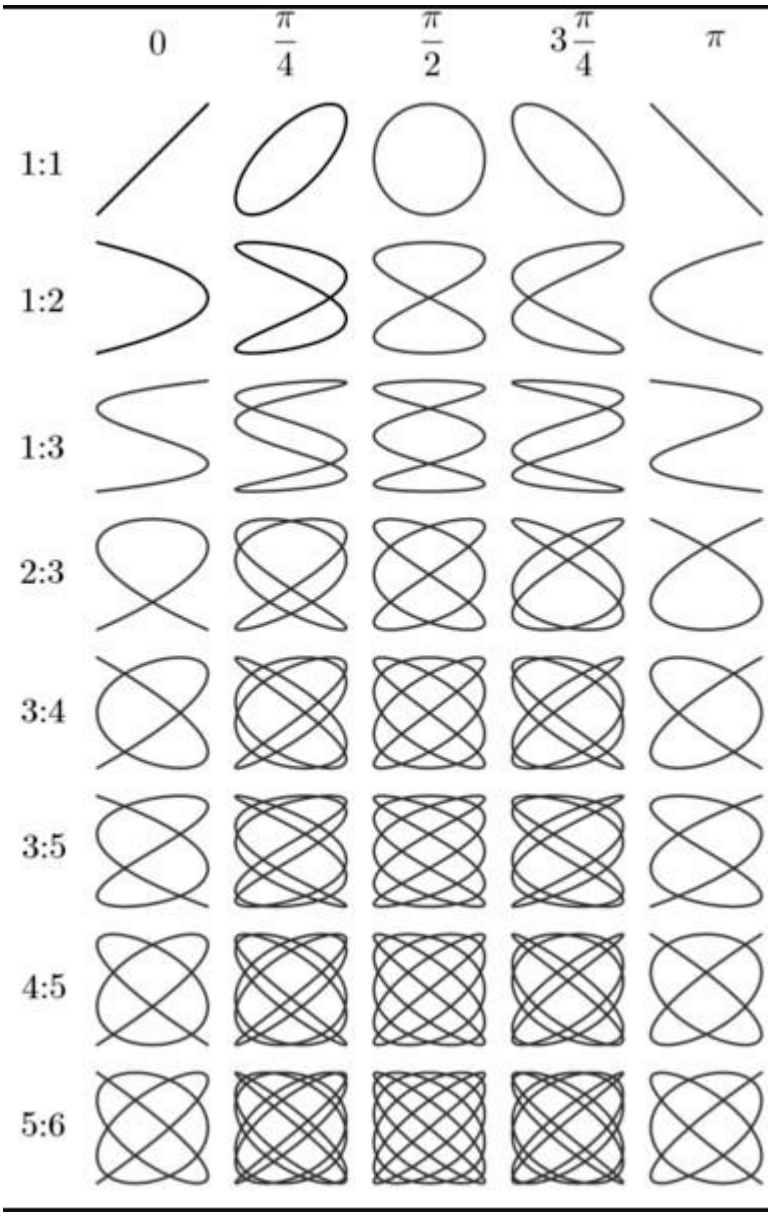


Figure 3-1: Lissajous patterns with phase and magnitude changes

Consider two sinusoids that form a Lissajous pattern, x(t) = A1sin(1t + 1) and y(t) = A2sin(2t + 2). The difference between phase angles 1 and 2 represents the phase angle relationship between the two sinusoids.

- If $\theta 1 = \theta 2$, the two sinusoids are in phase, and the Lissajous pattern will be a straight line that oscillates along the either x-axis or y-axis, and it also depends on the amplitude and frequency of the sinusoids.
- If $\theta 1 = \theta 2 + 90$ degrees, the two sinusoids are orthogonal, and the Lissajous pattern is a symmetric circle about the origin.
- If $\theta 1 = \theta 2 + 180$ degrees, the two sinusoids are 180 degrees out of phase, and the Lissajous pattern is a straight line that oscillates along the x-axis or y-axis, but in the opposite direction of the other sinusoids. r case.

Observing the ratio of the amplitudes, when the ratio of the amplitudes A1 and A2 gives the magnitude relationship between the two sinusoids.

- If A1 = A2, the two sinusoids have the same amplitude, and the Lissajous pattern will be symmetric about the origin.
- If $A1 \neq A2$, the two sinusoids have different amplitudes and the Lissajous pattern will be elongated or compressed in the x- or y-direction.

2. Generating a square pulse by using a saw tooth waveform and the effect of changing the potentiometer connected to pin 2 of IC741.

A square pulse can be generated by applying a saw tooth pulse and a reference voltage to an LM741 IC-based comparator circuit. The saw tooth waveform is generated using a signal generator, and the reference voltage is generated using a DC power supply.

The addition of the potentiometer creates a voltage divider, allowing the reference voltage to be varied. The potentiometer is connected to LM741 IC pin 2.

Once the reference voltage and saw tooth waveform are established, the parts of the wave that are less than the reference voltage will have the negative saturation voltage value, while the parts that are greater than the reference voltage will have the positive saturation voltage value. This results in a square wave.

When the reference voltage changes, the ratio of time when the voltage of the saw tooth wave is greater than the reference and less than the reference changes. This alters the nature of the square waveform, resulting in a waveform with a different duty cycle.

4 References

[1] Tutorialspoint, "Lissajous Figures," July 2013. [Online]. Available: https://www.tutorialspoint.com/electronic_measuring_instruments/electronic_measuring_instruments_lissajous_figures.htm. [Accessed 05 December 2023].

[2] Dutta Promeet, "Lissajous figure," Britannica, 19 May 2009. [Online]. Available: https://www.britannica.com/science/Lissajous-figure. [Accessed 05 December 2023].